SPML Summer 2023

Generative AI

July 18, 2025

1 Signal Processing and Machine Learning (SPML) - Summer 2023

The new version of the course moves from an introduction to everything about Machine Learning to Generative Modeling or Generative AI in Signal Processing.

2 Tentative Outline

- **Part I – Estimation and Regression**
- **Part II – Generative Modeling**
  - Generative Adversarial Networks
  - Narrowing Flows
  - Gaussian Mixture Models
  - Variational Autoregressive Models
  - Diffusion Models
- **Part III – Applications**
  Application in Parameter Estimation, e.g., in Wireless Communications.

3 Part I – Estimation and Regression

3.1 What is the difference between estimation and regression?

In both estimation and regression, we aim to discover the relationship between variables. However:

- **Estimation** is solely based on observations of random variables.
- **Regression** requires observations of pairs of dependent and independent random variables.

4 1. Estimation

4.1 Overview

Consider the process:

$$x \ \rightarrow \ \text{Estimator} \ \rightarrow \ y = 2 \tag{1}$$

Statistical models are often represented via explicit models, with the joint density function f_{XY} characterized based on data samples:

$$\{(x_i, y_i)\} \in S, \quad |S| = s.$$

4.2 Statistical modeling

- The observed variable $x \in \mathbb{R}^m$, with $x_i \sim f_x(x_i) = \int_X f_{XY}(x_i, y)\, dy$
- The estimated variable (e.g., the parameter or response) $y \in \mathbb{R}^m$, with $y_i \sim f_{Y|X}(y_i|x)$

5 2. Estimation Techniques

5.1 Maximum Likelihood (ML) Estimation

When partial knowledge of the conditional density $f_{X|Y}(x|y)$ is available, the maximum likelihood estimate (MLE) is derived as:

$$\boxed{\hat{y}_{ML} = \arg\max_{y \in Y} \sum_{x \in Q} \log f_{X|Y}(x; y)} \tag{2}$$

assuming the observations $Q = \{x_1, x_2, \ldots, x_Q\}$ are independent and identically distributed (i.i.d.). This is further discussed in the Signal Processing (SSP) course.

More refined considerations involve the assumption:

$$X_i \sim f_{X_i}(x_i, y) \tag{3}$$

and the property that, asymptotically, the MLE estimators are efficient, converging to the Cramér-Rao lower bound for variance.

5.2 Conditional Mean Estimation (CME)

In the case of full statistical model knowledge, the conditional mean estimator (CME) is:

$$\boxed{\hat{Y}_{CME} = \mathbb{E}[Y|X_1 = x_1, \ldots, X_q = x_q] = \int y\, f_{Y|X_1..X_q}(y|x_1, \ldots, x_q)\, dy} \tag{4}$$

which equals the conditional expectation of Y given the observations.

6 3. Special Case: Gaussian Distribution

When X and Y are jointly Gaussian, let:

$$Z = \begin{bmatrix} X \\ Y \end{bmatrix} \in \mathbb{R}^{n+m}$$

with mean vector:

$$\boldsymbol{\mu}_Z = \begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{bmatrix}$$

and covariance matrix:

$$\mathbf{Q}_z = \begin{bmatrix} C_x & C_{xy} \\ C_{yx} & C_y \end{bmatrix}$$

where:

$$C_x = \mathrm{Var}[X] = \mathbb{E}[(X - \boldsymbol{\mu}_x)(X - \boldsymbol{\mu}_x)]$$
$$C_y = \mathrm{Var}[Y] = \mathbb{E}[(Y - \boldsymbol{\mu}_y)(Y - \boldsymbol{\mu}_y)]$$
$$C_{xy} = \mathrm{Cov}[X, Y] = \mathbb{E}[(X - \boldsymbol{\mu}_x)(Y - \boldsymbol{\mu}_y)]$$

with symmetry $C_{xy} = C_{yx}$.

The linear minimum mean square error (LMMSE) estimator for Y given X is:

$$\boxed{\hat{Y}_{LMMSE} = C_{yx}C_x^{-1}(x - \boldsymbol{\mu}_x) + \boldsymbol{\mu}_y} \tag{5}$$

which minimizes the mean squared error (MSE):

$$\text{MSE} = \text{trace}\left[C_y - C_{yx}C_x^{-1}C_{xy}\right] \tag{6}$$

7 4. Conditional Prior

In many applications, it is assumed that the prior distribution of Y conditioned on X, denoted $f_{Y|X}$, is available as a mixture:

$$f_{Y|X}(y|x) = \int f_{Y|X,\theta}(y|x,\theta) f_\theta(\theta)\, d\theta \tag{7}$$

where $f_\theta(\theta)$ is a prior over the hyper-parameters, and $f_{Y|X,\theta}(y|x,\theta)$ is a feasible conditional prior.

In such cases, the conditional expectation $E_{X|Y}[Y|X]$ can be expressed as:

$$E_{X|Y}[Y|X] = E_{X|Y,\theta}[Y|X,\theta]\big|_{X=x} \tag{8}$$

which involves integrating over the hyper-parameters:

$$Y_{CME} = E_{d|x}\left[\int_Y y\, f_{Y|X,\delta}(y|x,\delta)\, dy\, x\right] \tag{9}$$

8 5. Summary of Estimator Types

- The **Conditional Mean Estimator** (CME) minimizes the mean squared error:

$$\hat{Y}_{MSE} = \arg\min_{Y \in \mathcal{Y}} \mathbb{E}_{Y_x}\left[(Y - \hat{Y})^2 \mid x\right] \tag{10}$$

  with

$$\mathbb{E}_{Y_x}\left[(Y - \hat{Y})^2 \mid x\right] = \int_Y (Y - \hat{Y})^2 f_{Y_x}(Y|x)\, dY \tag{11}$$

- An example for the case of linear estimators involves averaging over linear estimators:

$$Y_{ME} = \mathbb{E}_X\left[\frac{Y^{(1)}_{LMSSE}(x,s)}{x}\right] \tag{12}$$

Note

Further details, derivations, and theorems related to estimation theory, such as the Cramér-Rao bound and properties of estimators, are covered in the Signal Processing (SSP) course.